Student Academic Performance Prediction

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1. Abstract

The current document is a report related to student performance prediction. The objective of this model is to predict the quality of academic performance, whether high, average, or poor, depending on the behavior patterns of students. It is noted by experts that the source of information utilized includes a total number of 3,000 learners in the field of education at Kaggle, where there are 30 variables involved, including learning time, concentration level, procrastination level, and consistency rate. All processes were implemented through Python on Google Colab.

Testing with a decision tree classifier reached a test accuracy of 51.83% on purely behavioral data when exam scores were not used. It has been observed that procrastination index, consistency score, study hours, and focus score appear as the strongest indicators for academic performance. Because class imbalance existed, the analytical framework was unable to identify the high class well since only 14.8% of educational learners belonged to the high performers.

2. Introduction

The current project is about the prediction of academic achievement, and it is one of the areas where much research has been conducted to assist educational institutions in identifying their struggling students.

In my case, I have designed a machine learning pipeline that uses multiple factors such as hours of study, attendance, socio-economic factors, etc., in its prediction process. All of my code is written in Python; Pandas and scikit-learn are libraries I have used here.

3. Project Objective

The main objective of this project:

- Loading and exploring the dataset through Pandas.
- Use of imputation to fill in the missing data
- Numerical and categorical variables by Label Encoding
- Splitting and scaling for model building
- Decision Tree Classifier for multiclass classification
- And finally, modeling the same

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4. Methodology

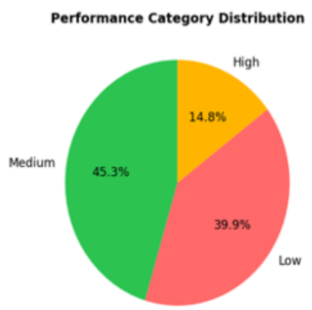
The whole process has 10 steps

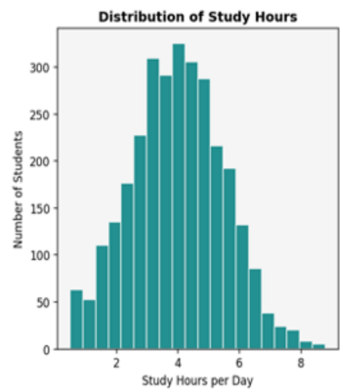
Step	Action	Tool Used
1	Load dataset from CSV	pandas — pd.read_csv()
2	Identify numerical and categorical columns	pandas — select_dtypes()
3	Check for missing values	pandas—isnull(). sum()
4	Fill missing values (median for numbers, mode for categories)	pandas — fillna()
5	Convert word columns to numbers (e.g., Male = 1, Female = 0).	scikit-learn — LabelEncoder
6	Separate input features (X) and target variable (y)	pandas — drop()
7	Split data — 80% train, 20% test	scikit-learn — train_test_split
8	Scale all features to range 0–1	scikit-learn — MinMaxScaler
9	Train the Decision Tree and measure accuracy	scikit-learn — DecisionTreeClassifier
10	Extract and rank feature importances	model.feature_importances_

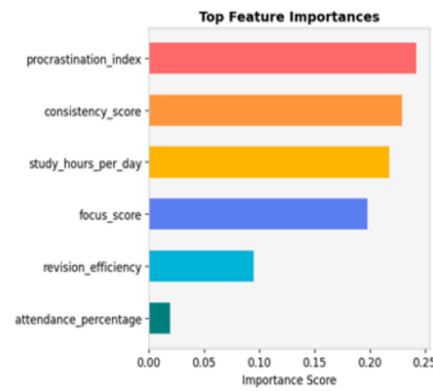
5. Data Analysis and Results

Metric	Value
Total Students	3,000
Total Features	30 behavioural attributes
High Performers	444 (14.8%)
Medium Performers	1,360 (45.3%)

Low Performers	1,196 (39.9%)
Female Students	1,528 (50.9%)
Male Students	1,472 (49.1%)







Metric	**Value**
Test Accuracy	51.83%
Training Set	2,400 students (80%)
Test Set	600 students (20%)
Best Predicted Class	Low (F1 = 0.57)
Worst Predicted Class	High (F1 = 0.00 — class imbalance)

Feature	**Importance**	**What it means**
Procrastination Index	24.2%	Students who delay work tend to perform worse
Consistency Score	22.9%	Regular study habits strongly predict better outcomes
Study Hours per Day	21.7%	More study time generally means better performance
Focus Score	19.8%	The quality of attention during study is nearly as important as hours
Revision Efficiency	9.5%	How well students revise has a moderate influence

6. Conclusion

In my project, I employed machine learning techniques to solve the problem in question. Predicting the success rate of the students at schools depending on their everyday behavior. I utilized the data set of 3000 students and made a decision tree to categorize the results of their activities into high, medium, and low.

The results of the prediction were 51.83% accurate. Behavior is multi-dimensional; thus, it is difficult to predict the outcomes according to the specific habit, such as the amount of time spent on learning and concentration. To predict something with an accuracy of 50 percent using behavioral indicators is already a success.

From the analysis, we learn that the first predictor is procrastination, followed by consistent studying, daily hours of studying, and quality of concentration. This indicates that the matter is not only in hours spent studying but also in regularity and diligence. It was surprising that factors like social media use, gaming time, and sleep had no impact on the data set. In essence, it appears that the activities that students engage in during their study sessions have more impact on their grades than other activities. The decision tree algorithm used here is relatively easy to comprehend. Essentially, it follows a process where it poses a series of 'yes' or 'no' questions till a conclusion is made. This means that the model used here is a transparent machine learning model that is valuable in the education sector.

APPENDICES

Appendix A — References

- Kaggle Dataset: https://www.kaggle.com/datasets/aiexplorer77/academic-performance-prediction
- scikit-learn: https://scikit-learn.org/stable/
- pandas: https://pandas.pydata.org/docs/
- Google Colaboratory: https://colab.research.google.com/

Appendix B — GitHub Link

- https://github.com/oishimaity2024-web/student-performance-prediction

Appendix C — Document Link

- https://colab.research.google.com/drive/1eLgPP7GJyN0IPJWzSsXUZ6_uTHtM8iPs?usp=sharin

- https://docs.google.com/document/d/1S222kxMAnjq0sOnyGbqrRZe2xRQZk9k5qwZO30VpIvs/edit?usp=drive_link